Research Assignment 2.

Foundational concepts of Analytics

① Business Analytics is the process of using data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help management gain insights about business operations and make fact based and data driven conclusions

② Data Analytics is the science of analysing raw data to find patterns, identify trends and draw conclusions about that information

③ Business Intelligence refers to the procedural and technical infrastructure that collects, stores and analyze the data produced by a company's ~~entity~~ activities. It encompasses a combination of strategies, tools, and methodologies aimed at transferring raw data into meaningful, actionable insights for tactical and strategic decision making

④ Data driven decision making is the process of using facts, metrics and data analysis to guide strategic business decisions rather than relying soley on intuition, gut feeling or observation. to minimise risks and maximise business impact

⑤ Big data refers to datasets with high volume, velocity and variety that they cannot be processed or managed by traditional database software within a reasonable time. It represents a shift from analysing data stocks (historical records) to analysing data flows (real time data) to generate actionable insights and competitive advantage

(6) Descriptive analytics is the foundational process of using historical and current data to identify trends, patterns and relationships answering the fundamental question that is "what happened".

(7) Diagnostic Analytics is a form of advanced data analysis that examines historical data to understand the "why" behind past trends, anomalies or performance outcomes.

(8) Predictive Analytics is a branch of advanced analytics that uses historical data, statistical modelling, data mining techniques and machine learning to identify the liklihood of future outcomes.

(9) Prescriptive analysis is the advanced stage of business analytics that uses AI and mathematical modelling to ~~explain~~ recommend the best actionable strategies, answering "what should we do next". By analysing data to optimize outcomes, this approach moves beyond prediction to recommend specific actions to improve performance such as price optimisation or resource allocation.

(10) Data literacy is the ability to read, work with, analyze and communicate with data to foster a data driven culture and make informed decisions. It is considered a crucial "thinking skill" rather than a requirement to become a data scientist, enabling non-technical professionals to turn raw numbers into meaningful insights.

4 pillars: Reading, working with, analysing and communicating data

Data, measurement & Reporting

(11) Structured data refers to information that is highly organized, formatted and adheres to a rigid, predefined schema. It is typically quantitative, making it easily searchable, retrievable and analyzable by both humans and machines.

(12) Unstructured data is information that does not have a pre-defined data model, schema, or organisational structure. Unstructured data is generally qualitative, text heavy or multimedia based, making it difficult to search, store and analyze using conventional relational databases

(13) Data warehouse is a centralized repository that stores current and historical data from multiple sources such as ERP & CRM & transactional systems, which are designed to enable analysis, reporting and data driven decision making. It serves as a single source of truth for an organisation, acting as a "database of databases" to support business intelligence (BI) activities

(14) Data mining is the practice of automatically searching large stores of data to discover patterns and trends that go beyond simple analysis. It is often used as a synonym for "knowledge discovery in data" (KDD) and uses methods like classification, clustering and predictive analysis to make data actionable

(15) Data visualisation is the strategic process of turning raw data into visual formats → such as charts, graphs, maps and dashboards to make complex information, relationships, and trends instantly understandable, accurate, and actionable for decision makers.

(16) Dashboards are a visual interface that aggregates, organises and displays key metrics, data points and Key Performance Indicators (KPIs) on a single screen. Its purpose is to provide a real time or near real time snapshot of operational, strategic or analytical data, enabling users to monitor performance, identify trends, and make faster, data driven decisions

(17) A Metric is a quantifiable, numerical measure used to track and assess the status of a specific business process. Metrics are generally operational and tactical, providing data on

(18) A Key Performance Indicator is a specialised, high level metric that directly measures how well an organisation is achieving its most critical, strategic, or operational objectives. KPIs are a subset of metrics.

(19) Benchmark is a standard or point of reference against which an organisation's performance, processes or products can be compared and measured. It is a critical tool for data driven decision making, allowing companies to evaluate whether their performance is strong, weak or consistent with industry norms, competitors or internal historical data.

(20) Segmentation is a strategy in which businesses categorise potential customers into distinct groups based on shared characteristics such as demographics, behaviours, geography or psychographics, to tailor products and marketing more effectively

Customer & Sales KPIs

(21) Conversion rate is the percentage of users or visitors who complete a specific, desires action out of the total number of visitors or potential prospects

(22) Customer Acquisition cost (CAC) is the total expense a business incurs to persuade a potential customer to buy a product or service.

The formula = $\frac{\text{total sales and marketing expenses}}{\text{number of new customers acquired}}$

(23) Customer Lifetime Value is a metric that defines the total revenue or profit a business can expect to earn from a single customer account throughout their entire relationship. It moves beyond measuring individual transactions to assess the long term value of acquiring and retaining customer

(24) Churn rate is a key performance indicator (KPI) that measures the percentage of customers, subscribers or employees who stop doing business with an entity over a specific period of time.

(25) Customer Retention Rate measures the percentage of existing customers who remain loyal to a business over a specific period, reflecting the organisations ability to prevent churn and encourage repeat purchases.

(26) Net promoter score (NPS) is a standardised key performance indicator that measures customer loyalty, satisfaction and enthusiasm towards a company, product or service

(27) Average Order Value (AOV) tracks the average amount spent each time a customer places an order on a website or mobile app. It is a key performance indicator used to evaluate the effectiveness of an ecommerce store's marketing, pricing and merchandising strategies

$$\text{Formula} = \frac{\text{Total revenue}}{\text{Total number of orders}}$$

(28) Sales funnel is a strategic step by step framework that visualises the customer journey from initial brand awareness to the final purchase decision. It acts as a marketing and sales model to guide prospects through distinct stages, with the funnel shape representing the reduction in the number of potential customers (leads) as they move closer to closing a deal.

Marketing & Campaign Analytics

(29) A marketing campaign is a strategic, organised and time bound effort designed to promote a specific business goal such as a product launch, increased brand awareness, or lead generation by coordinating messages and tactics accross multiple channels

(30) A post campaign analysis is the systematic, data driven evaluation of a marketing initiative after its conclusion to measure effectiveness, calculate ROI and derive actionable insights for future strategies. It involves reviewing key performance indicators against initial goals to determine what worked, what failed and why

(31) Incremental revenue is the additional income a company generates from a specific business activity, decision or marketing initiative beyond the baseline revenue it would have earned otherwise. It is used to measure the true lift or financial impact of an action such as launching a new product by isolating it from natural organic growth or seasonality

(32) Return on Investment (ROI) is a performance measure used to evaluate the efficiency or profitability of an investment relative to its cost. Expressed typically as a percentage, ROI helps business leaders compare the effectiveness of different investments and make data driven decisions.

(33) Return on Ad Spend (ROAS) is a marketing metric used to measure gross revenue generated for every amount spent on advertising. It serves as a KPI to evaluate campaign efficiency, focused on top-line revenue rather than net profit

(34) Click Through Rate (CTR) measures the percentage of ad viewers who click an advertisement compared to the total number of ad impressions.

(35) A bounce rate is the percentage of sessions that were not engaged representing the opposite of engagement rate. It is basically the percentage of visitors who land on a page and leave without taking any further action or visiting other pages.

(36) A/B Testing is a scientific, randomised controlled experiment used to compare two or more versions of a digital asset such as a webpage, email, app feature or advertisement to determine which performs better based on specific key performance indicators (KPIs)

(37) Lead generation is the initiation of consumer interest or inquiry into the products or services of a business. It is the process of identifying, attracting and converting strangers and prospects into someone who has indicated an interest in a company's product or service.

(38) Marketing attribution is the analytical process of identifying which marketing touchpoints such as ads, emails or social media posts contribute to a desired customer action and assigning credit to those interactions.